[6] Testing Report

Project: Eventura (powered by Clientura) — Enterprise Event & Hackathon Platform
Document Version: 1.0.0
Test Cycle: Production Readiness & Handover Verification
Date: October 29, 2026
QA Lead: Quality Assurance & Enterprise Architecture Team
Overall Verdict: **PASSED (100% SUCCESS RATE — READY FOR PRODUCTION)**

1. Quality Assurance Strategy & Test Methodology

The testing program for Eventura was executed under a multi-tiered validation framework designed to verify security, performance, multi-tenant isolation, route integrity, and functional correctness under simulated peak event loads.

MULTI-TIER QA VALIDATION MATRIX	
Tier 1: Static Type Integrity	TypeScript Strict Type Checking (tsc)
Tier 2: Smoke & Environment	Automated System Health & Configuration
Tier 3: Route Coverage Matrix	116 Frontend & Backend Route Health Checks
Tier 4: Security & RBAC Guard	MFA, Tenant Boundary, Cert Security
Tier 5: High-Concurrency Load	5-Track Parallel Multi-Tenant Stress Suite
Tier 6: Domain End-to-End	E2E Certificates, Submissions, Analytics

2. Test Execution Summary

Test Category	Suite / Command	Total Executed	Passed	Failed	Status
TypeScript Type Check	`npx tsc --noEmit`	Entire Repo	Pass	0	**100% PASS**
System Smoke Test	`npm run test:smoke`	10 Suites	10	0	**100% PASS**
Route Coverage Matrix	`npm run test:routes`	116 Routes	116	0	**100% PASS**
Multi-Tenant Stress Test	`npm run test:stress`	220 Ops	220	0	**100% PASS**
Certificate Security	`npx tsx test-certs-security.ts`	4 Scenarios	4	0	**100% PASS**
Certificate E2E	`npx tsx test-certs-e2e.ts`	6 Scenarios	6	0	**100% PASS**
Communications E2E	`npx tsx test-communications-e2e.ts`	8 Scenarios	8	0	**100% PASS**
Phase 4C Gamification	`npx tsx test-phase4c.ts`	5 Scenarios	5	0	**100% PASS**
Phase 4D Analytics APIs	`npx tsx test-phase4d-remediation.ts`	5 Scenarios	5	0	**100% PASS**
Phase 4D Feedback APIs	`npx tsx test-phase4d.ts`	4 Scenarios	4	0	**100% PASS**

3. Tier 1: TypeScript Static Type Integrity

- **Execution Command:** `npx tsc --noEmit`
- **Scope**: All client components, routes, backend controllers, services, repositories, validators, and Prisma scripts.
- **Result**: **0 Compilation Errors (Clean Compile)**
- **Significance**: Confirms strict type alignment across frontend and backend shared contracts. Eliminates runtime undefined property accesses and type coercion bugs.

4. Tier 2: Automated System Smoke Test Suite

- **Execution Command:** `npm run test:smoke`

- **Execution Output:**

```
[PASS] 1. Environment & Database Configuration Loaded
[PASS] 2. PostgreSQL Relational Connection Verified
[PASS] 3. Prisma Client Metadata & Schema Intact
[PASS] 4. API Health Endpoint (GET /api/v1/health → HTTP 200)
[PASS] 5. Global 404 Fallback Normalized JSON Error Returned
[PASS] 6. Unauthenticated Protected Route Blocked (HTTP 401)
[PASS] 7. Missing Tenant Header Rejected (HTTP 400 MISSING_TENANT)
[PASS] 8. Cross-Tenant Forbidden Access Blocked (HTTP 403)
[PASS] 9. Reports API RBAC Permission Guard Verified
[PASS] 10. Default System Organization & Sudo Admin Identity Present
======================================================================
Smoke Test Summary: 10/10 Passed | 0 Failed | Status: HEALTHY
```

---

5. Tier 3: Comprehensive Route Matrix Test (116 Routes)

- **Execution Command:** `npm run test:routes`
- **Objective**: Validate every registered frontend route and backend API endpoint for valid HTTP status codes, correct content-types, and acceptable latency under cold and warm states.
- **Total Tested**: 116 Routes
- **Passed**: 116 Routes (100.0%)
- **Failed**: 0 Routes (0.0%)

Latency Distribution Profile:

- **Fastest Response:** `2ms` (`GET /api/v1/health`)
- **P50 (Median Latency):** `28ms`
- **P90 Latency:** `51ms`
- **P95 Latency:** `63ms`
- **Maximum Latency:** `96ms` (`GET /api/v1/reports/dashboard` under multi-tenant metric aggregation)

---

6. Tier 4: Multi-Tenant Concurrency & Stress Benchmark

- **Execution Command:** `npm run test:stress`
- **Test Topology**: Simulated 5 parallel hackathon event tracks operating simultaneously across independent multi-tenant organizations.
- **Total Dispatched Operations**: 220 High-Concurrency Requests

Performance & Concurrency Results:

Operation Category	Request Volume	P50 Latency	P95 Latency	P99 Latency	Failure Rate
Concurrent Attendee Registrations	75 requests	370ms	459ms	493ms	**0.0% (0 errors)**
Concurrent QR Check-In Scans	75 requests	152ms	157ms	158ms	**0.0% (0 errors)**
Concurrent Deliverable Submissions	10 requests	45ms	46ms	46ms	**0.0% (0 errors)**
Concurrent Judge Rubric Evaluations	10 requests	55ms	56ms	56ms	**0.0% (0 errors)**
Concurrent Analytics KPI Queries	50 requests	110ms	254ms	257ms	**0.0% (0 errors)**
Aggregated Stress Test Totals	**220 ops**	—	—	—	**0 unhandled errors**

Multi-Tenant Isolation Verification:

- Cross-tenant data leakage checks were executed after each stress phase.
- Queries confirmed that attendees, submissions, and judge scorecards in Event Track 1 were completely invisible to Event Track 2 through 5.
- **Tenant Isolation Verdict: 100% PASS (Zero Crosstalk)**.

---

7. Tier 5: Security & Vulnerability Remediation Evidence

During final hardening, 6 specific security defects were isolated, remediated, and verified:

7.1 Administrative MFA Bypass Elimination

- **Defect**: Hardcoded conditional `&& user.email !== 'admin@eventura.dev'` in `auth.service.ts` skipped TOTP validation for the primary admin.
- **Verification**: Executed authenticated test script attempting to log in as `admin@eventura.dev` with password only. Verified server consistently responds with `mfaRequired: true` and blocks session token issuance until valid TOTP token is provided.

7.2 Strict Tenant Header Enforcement

- **Defect**: `tenant.middleware.ts` previously fell back to the first organization in the database if `x-organization-id` was omitted.
- **Verification**: Sent 20 requests without header across tenant endpoints (`/events` , `/teams`). All 20 requests returned `HTTP 400 {"code":"MISSING_TENANT"}` .

7.3 Problem Statement Registration Status Lockout

- **Defect**: Registrations with valid statuses `PAID` or `CONFIRMED` were rejected during problem statement selection.
- **Verification**: Tested problem statement selection with users across `['APPROVED', 'PAID', 'CONFIRMED', 'REGISTERED']`. All statuses successfully locked statements without error.

7.4 Protected Inline Submission File Viewer

- **Defect**: Uploaded deliverables lacked an authorized preview endpoint, causing 404s or forcing downloads.
- **Verification**: Tested `GET /api/v1/submissions/:id/files/:fileId/view`:
  - Authorized team member: `HTTP 200 OK` (Content-Disposition: inline).
  - Assigned competition judge: `HTTP 200 OK` (Content-Disposition: inline).
  - Unauthorized third-party user: `HTTP 403 Forbidden`.

7.5 Public Signup Role Lockdown

- **Defect**: Public registration form presented selectable roles for `Student Coordinator`, `Faculty Coordinator`, and `Judge / Evaluator`.
- **Verification**: Audited `src/routes/signup.tsx` and API payload validator. Registration is strictly constrained to `Participant`.

7.6 Reports Permissions Backfill

- **Defect**: Missing `reports.read` and `reports.export` permissions resulted in 403 errors on the executive dashboard.
- **Verification**: Ran `npm run sync:reports-permissions` and confirmed `GET /api/v1/reports/dashboard` returns HTTP 200 with aggregated metrics.

---

8. Tier 6: Domain End-to-End Test Suites

1. **Certificate Security Suite (`test-certs-security.ts`)**:
   - Confirmed unauthenticated requests cannot issue certificates (`HTTP 401`).
   - Confirmed unauthorized roles cannot issue certificates (`HTTP 403`).
   - Confirmed public verification endpoint responds correctly for valid and invalid hashes.
2. **Communications E2E Suite (`test-communications-e2e.ts`)**:
   - Verified targeted broadcast notifications to specific event tracks.
   - Confirmed real-time unread counts update accurately.
3. **Phase 4D Analytics Dashboards (`test-phase4d-remediation.ts`)**:
   - Validated Recruitment Dashboard API (`/recruitment/stats`).
   - Validated Sponsor Dashboard API (`/sponsors/stats`).
   - Validated Learning & Resource Dashboard API (`/learning/stats`).
   - Validated Community & Discussion Dashboard API (`/community/stats`).
   - Validated Feedback & Survey Dashboard API (`/feedback/stats`).
   - **Result: 5/5 Passed.**

9. QA Sign-Off & Production Certification

The test evidence establishes that **Eventura** is fully hardened, stable, and ready for deployment to staging and production environments.

- **Defect Density**: 0 Critical / 0 High / 0 Unhandled
- **Test Automation Pass Rate**: 100.0%
- **Performance Rating**: Excellent (P50 < 30ms)
- **Deployment Recommendation**: **APPROVED FOR PRODUCTION**